



PHYS121 Integrated Science-Physics

W6T3 The 2nd law of Thermodynamics

References:

- [1] David Halliday, Jearl Walker, Resnick Jearl, 'Fundamentals of Physics', (Wiley, 2018)
 - [2] Doug Giancoli, 'Physics for Scientists and Engineers with modern physics', (Pearson, 2009)
 - [3] Hugh D. Young, Roger A. Freedman, 'University Physics with Modern Physics', (Pearson, 2012)
- And others specified when needed.



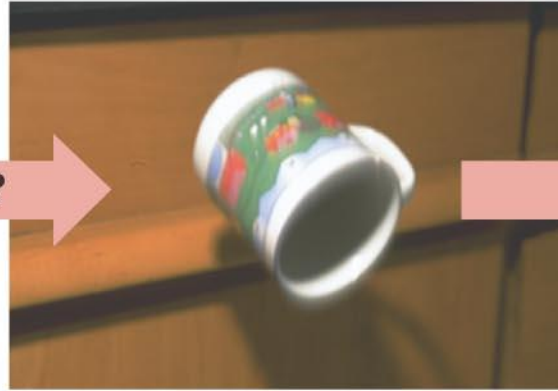


Learning Outcomes

- Explain the concept of entropy and calculate its value.
- Analyze how a typical heat engine works.
- Analyze how a typical refrigerator works.
- Explain the 2nd (statistic) and 3rd laws of thermal dynamics.



Initial state.



Later: cup reassembles
and rises up.

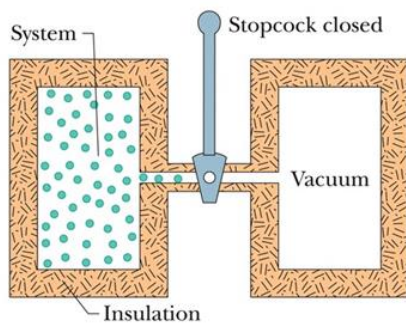


Later still: cup lands on table.

The first law of thermodynamics tells us that energy is conserved. However, the absence of the process illustrated above indicates that conservation of energy is not the whole story. If it were, movies run backwards would look perfectly normal to us!

Entropy

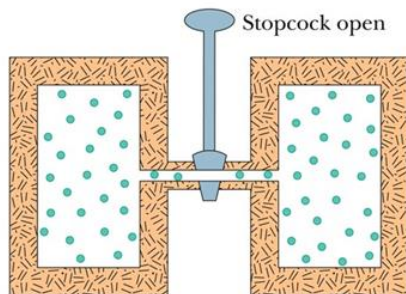
If an irreversible process occurs in a closed system, the entropy S of the system always increases; it never decreases.



(a) Initial state i



Irreversible process



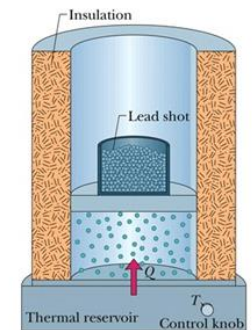
(b) Final state f

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This process is irreversible; that is, it does not occur in reverse, with the gas spontaneously collecting itself in the left half of the container.

The isothermal expansion of an ideal gas, done in a reversible way.

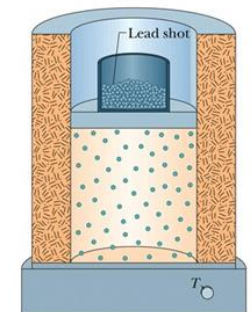
If a process occurs in a closed system, the entropy of the system **increases** for **irreversible** processes and remains **constant** for **reversible** processes. It never decreases.



(a) Initial state i



Reversible process



(b) Final state f

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Change in Entropy

$$\Delta S = S_f - S_i = \int_i^f \frac{dQ}{T}$$

$$\Delta S \geq 0 \quad (\text{second law of thermodynamics}),$$

Here Q is the energy transferred as heat to or from the system during the process, and T is the temperature of the system in kelvins during the process.

To find the entropy change for an irreversible process, replace that process with any reversible process that connects the same initial and final states. Calculate the entropy change for this reversible process with the above equation.

The Second Law of Thermodynamics

where the greater-than sign applies to irreversible processes and the equals sign to reversible processes. Equation applies only to closed systems.

$$\Delta S = S_f - S_i \approx \frac{Q}{T_{\text{avg}}}, \quad (20-3)$$

Example: Entropy change when mixing water.

A sample of 50.0 kg of water at 20.00°C is mixed with 50.0 kg of water at 24.00°C . Estimate the change in entropy.

The total entropy always increases when heat flows from a warmer object to a colder one in an isolated two-body system. The heat transferred is the same, and the cooler object is at a lower average temperature than the warmer one, so the entropy gained by the cooler one is always more than the entropy lost by the warmer one.

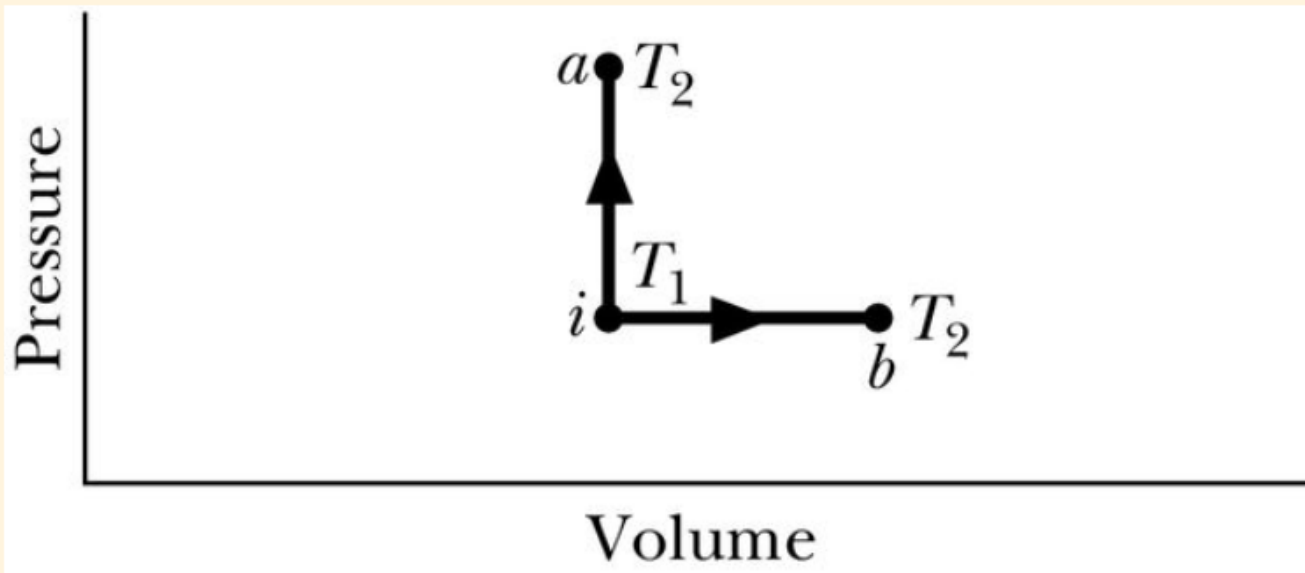
For an ideal gas goes through a reversible process:

$$\Delta S = S_f - S_i = nR \ln \frac{V_f}{V_i} + nC_V \ln \frac{T_f}{T_i}. \quad (20-4)$$



Checkpoint 2

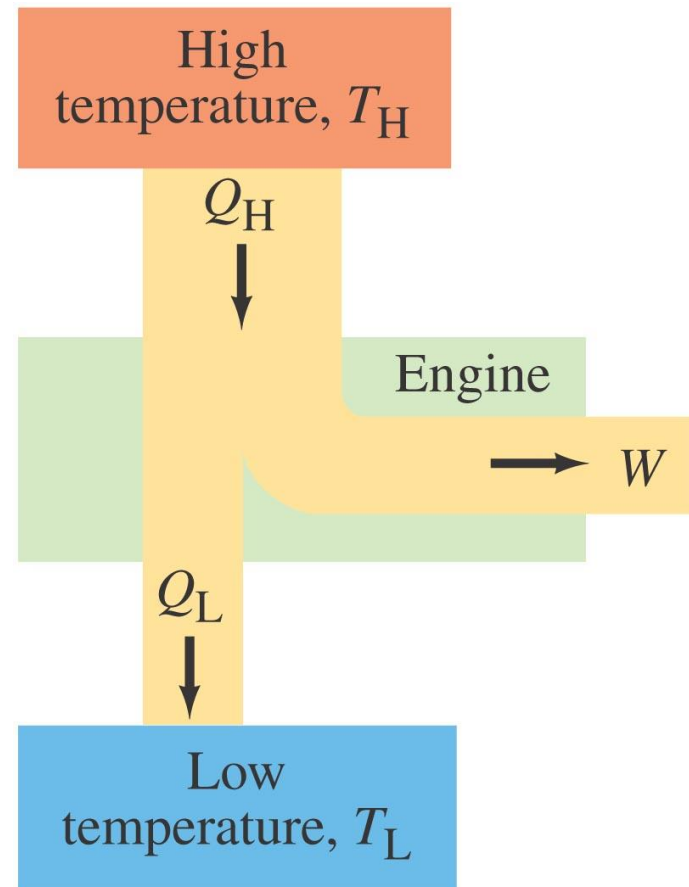
An ideal gas has temperature T_1 at the initial state i shown in the p - V diagram here. The gas has a higher temperature T_2 at final states a and b , which it can reach along the paths shown. Is the entropy change along the path to state a larger than, smaller than, or the same as that along the path to state b ?



Heat Engines

It is easy to produce thermal energy using work, but how does one produce work using thermal energy?

This is a heat engine; mechanical energy can be obtained from thermal energy only when heat can flow from a higher temperature to a lower temperature.

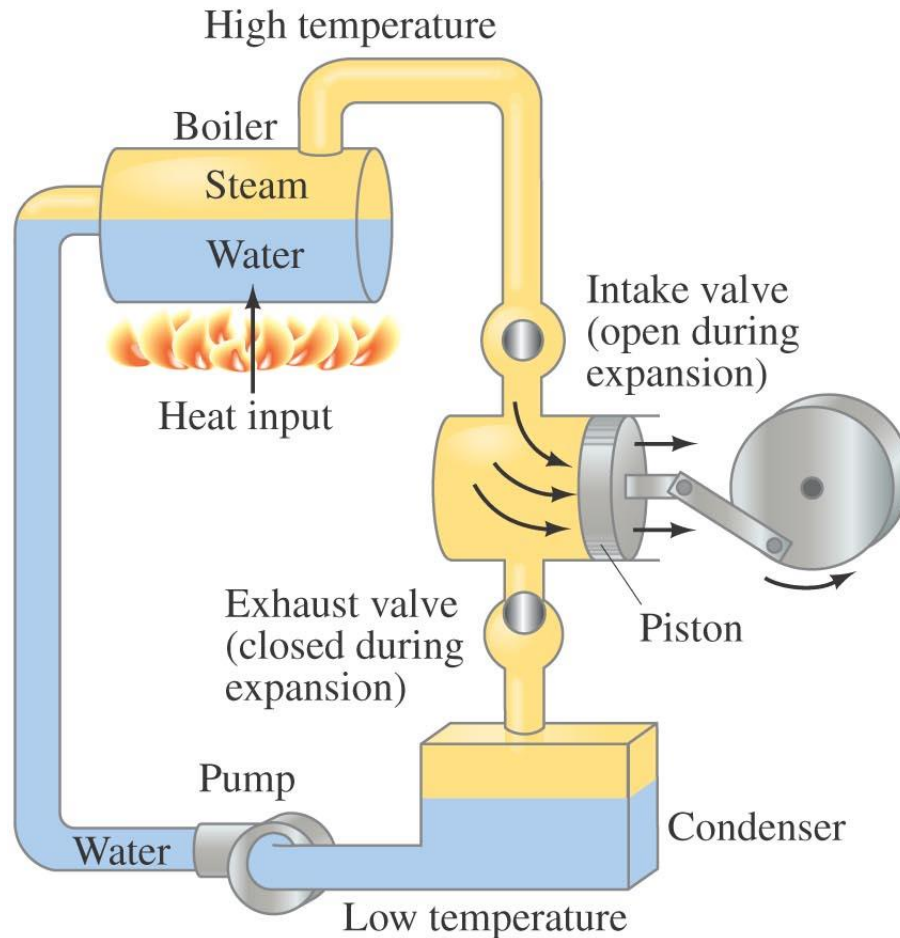


We will discuss only engines that run in a repeating cycle; the change in internal energy over a cycle is zero, as the system returns to its initial state.

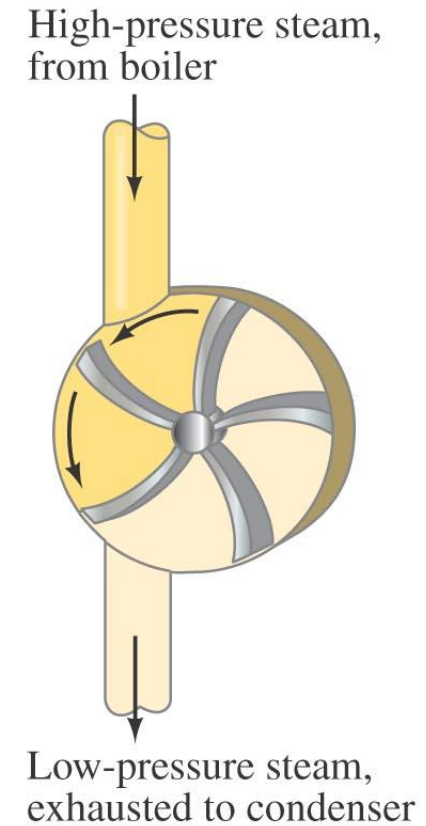
The high-temperature reservoir transfers an amount of heat Q_H to the engine, where part of it is transformed into work W and the rest, Q_L , is exhausted to the lower temperature reservoir. Note that all three of these quantities are positive.

A steam engine is one type of heat engine.

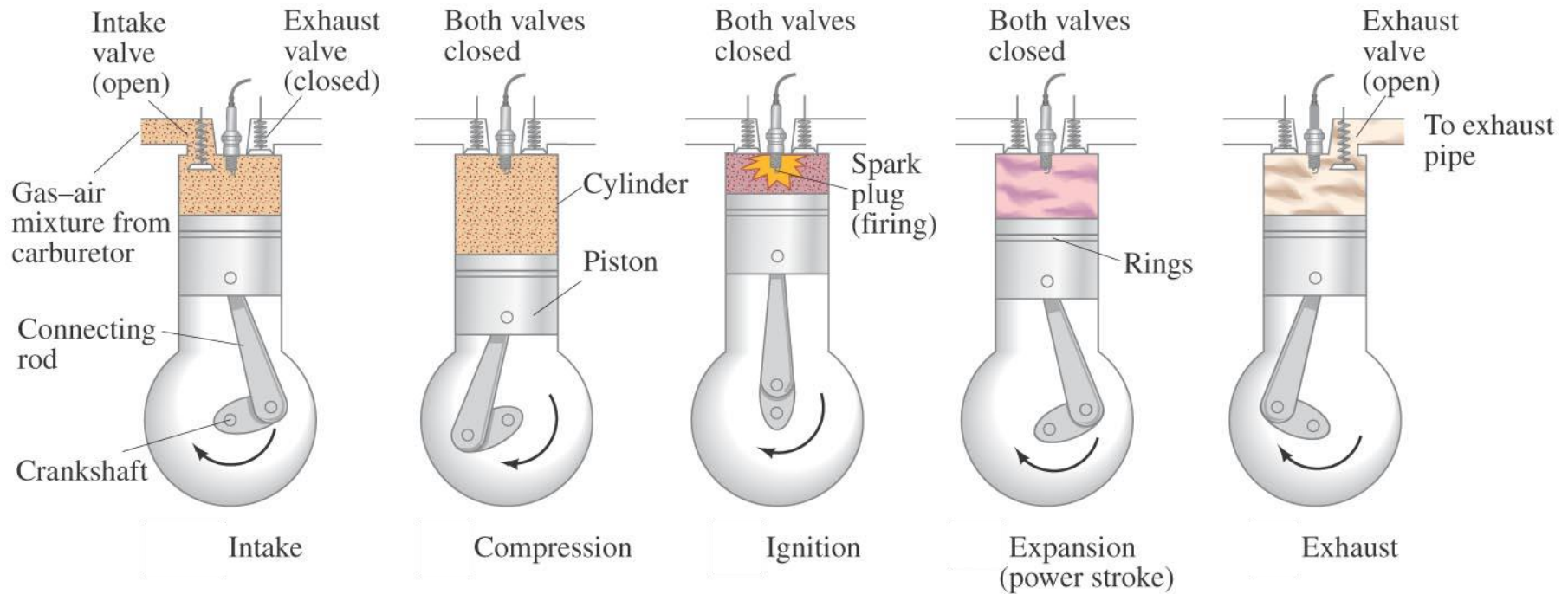
Reciprocating type



Turbine (boiler and condenser not shown)



The internal combustion engine is a type of heat engine as well.



Why does a heat engine need a temperature difference?

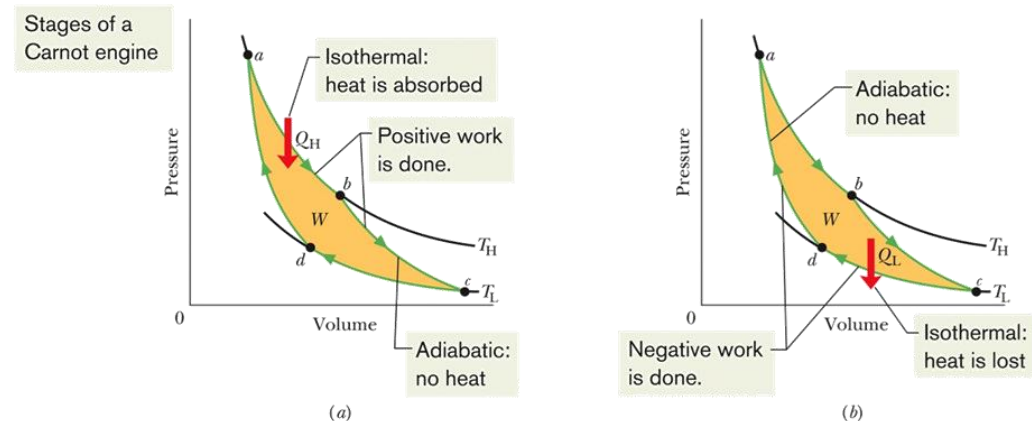
Otherwise the work done on the system in one part of the cycle would be equal to the work done by the system in another part, and the net work would be zero.

Entropy in the Real World: Engines

In an ideal engine, all processes are reversible and no wasteful energy transfers occur due to, say, friction and turbulence.

Carnot Engine

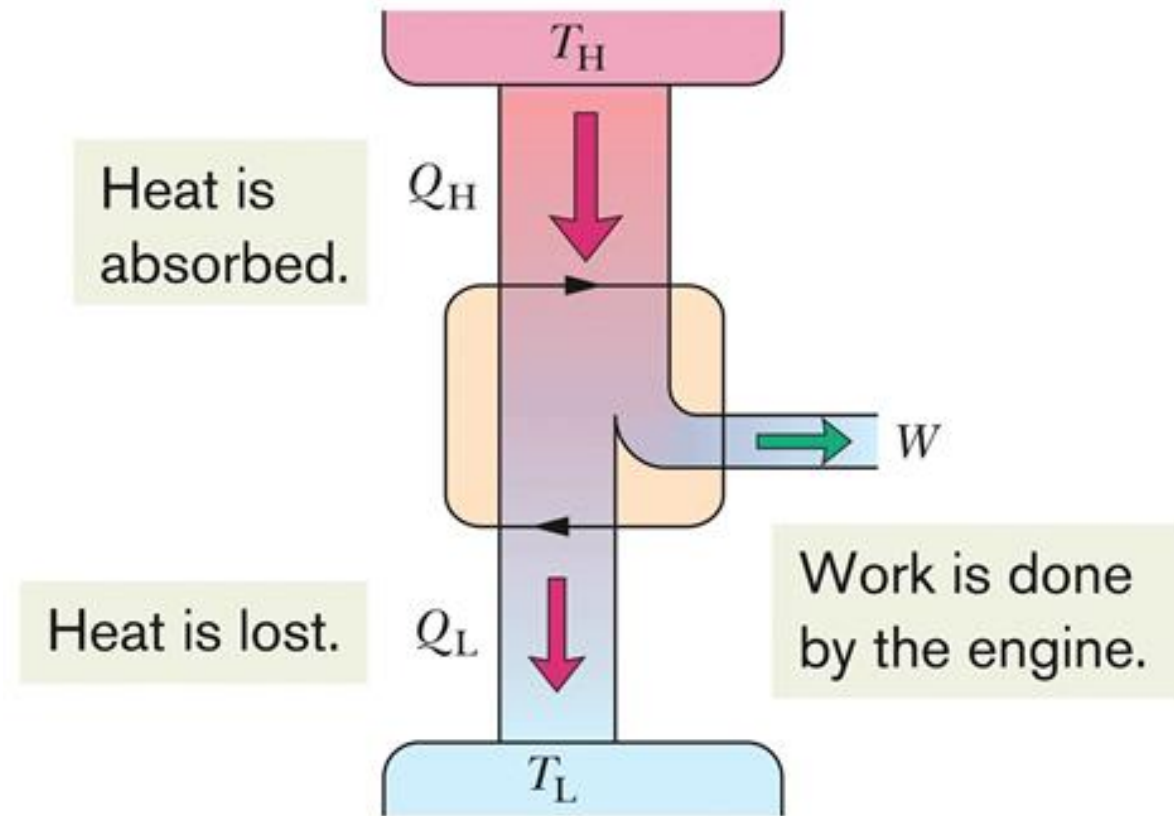
A pressure–volume plot of the cycle followed by the working substance of the Carnot engine. The cycle consists of two isothermal (ab and cd) and two adiabatic processes (bc and da). The shaded area enclosed by the cycle is equal to the work W per cycle done by the Carnot engine.



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Schematic of a Carnot engine

The elements of a Carnot engine. The two black arrowheads on the central loop suggest the working substance operating in a cycle, as if on a p - V plot.



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No series of processes is possible whose sole result is the transfer of energy as heat from a thermal reservoir and the complete conversion of this energy to work.

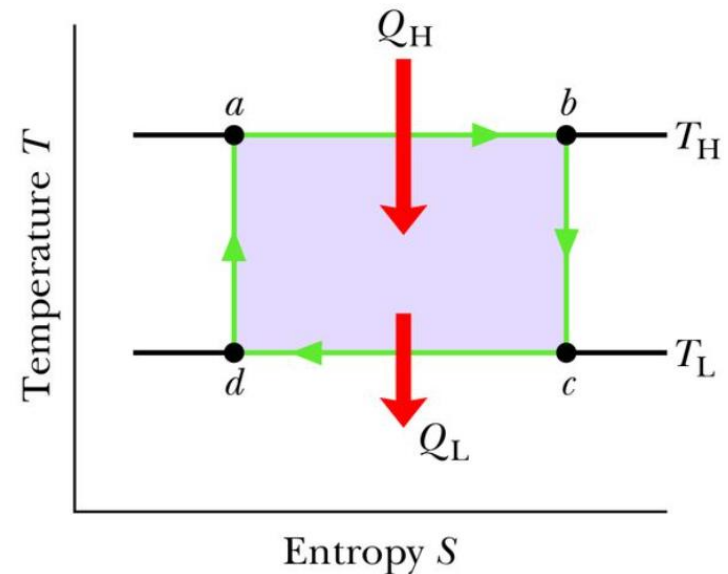
Efficiency of a Carnot Engine

Efficiency of any engine:

$$\varepsilon = \frac{\text{energy we get}}{\text{energy we pay for}} = \frac{|W|}{|Q_H|}$$

Efficiency of Carnot engine:

$$\varepsilon_C = 1 - \frac{T_L}{T_H}$$



Stirling Engine

The Stirling engine was developed in 1816 by Robert Stirling. This engine, long neglected, is now being developed for use in automobiles and spacecraft.

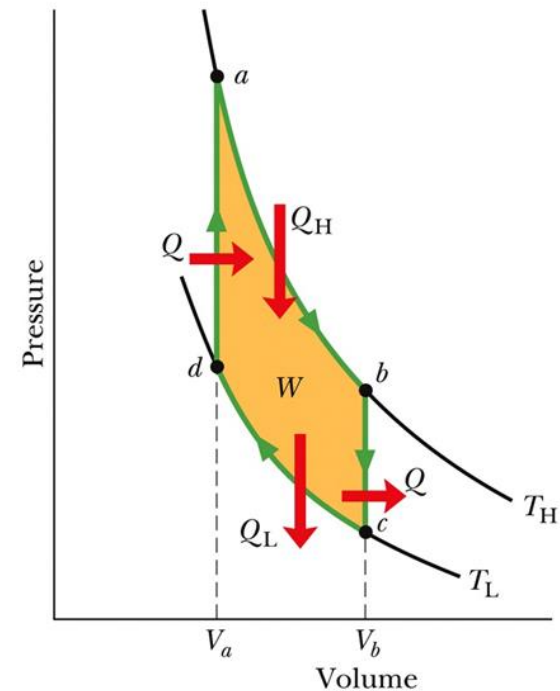
A p - V plot for the working substance of an ideal Stirling engine, with the working substance assumed for convenience to be an ideal gas.

Checkpoint 3

Three Carnot engines operate between reservoir temperatures of (a) 400 and 500 K, (b) 600 and 800 K, and (c) 400 and 600 K. Rank the engines according to their thermal efficiencies, greatest first.

Answer: (c), (b), (a).

Stages of a Stirling engine



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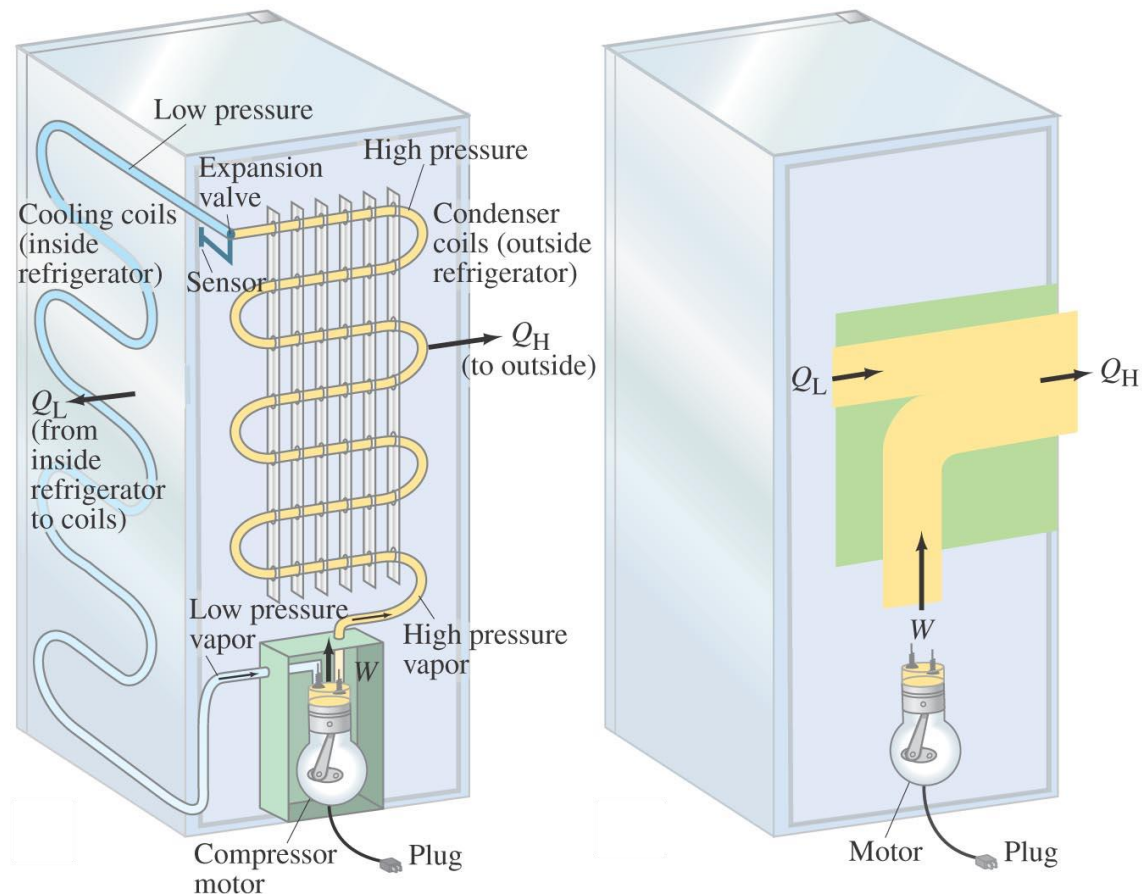
Example: Car efficiency.

An automobile engine has an efficiency of 20% and produces an average of 23,000 J of mechanical work per second during operation.

(a) How much heat input is required, and

(b) How much heat is discharged as waste heat from this engine, per second?

This figure shows more details of a typical refrigerator.



Refrigerator and Real Engines

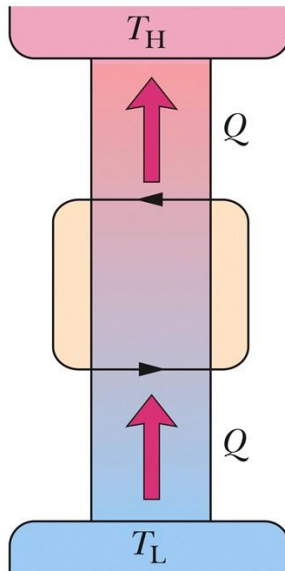
In an ideal refrigerator, all processes are reversible and no wasteful energy transfers occur as a result of, say, friction and turbulence.

Refrigerators

$$K = \frac{\text{what we want}}{\text{what we pay for}} = \frac{|Q_L|}{|W|} \quad (\text{coefficient of performance, any refrigerator}),$$

$$K_C = \frac{T_L}{T_H - T_L} \quad (\text{coefficient of performance, Carnot refrigerator}).$$

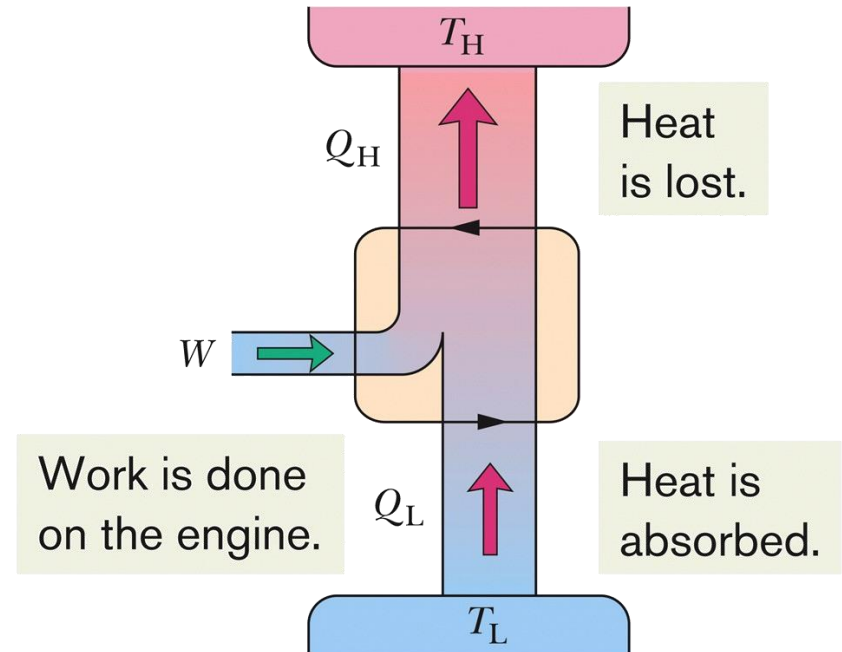
Perfect refrigerator:
total transfer of heat
from cold to hot
without any work



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The elements of a perfect refrigerator — that is, one that transfers energy from a low-temperature reservoir to a high-temperature reservoir without any input of work.

Schematic of
a refrigerator



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The elements of a refrigerator. Work W is done on the refrigerator (on the working substance) by something in the environment.

A Statistical View of Entropy

The **entropy** of a system can be defined in terms of the possible distributions of its molecules. For identical molecules, each possible distribution of molecules is called a **microstate of the system**. All equivalent microstates are grouped into a configuration of the system. The number of microstates in a configuration is the **multiplicity** W of the configuration.

For a system of N molecules that may be distributed between the two halves of a box, the multiplicity is given by

$$W = \frac{N!}{n_1!n_2!} \quad (\text{multiplicity of configuration}).$$

Here n_1 is the number of molecules in one half of the box and n_2 is the number in the other half. *A basic assumption of statistical mechanics is that all the microstates are equally probable.*

An insulated box contains six gas molecules. Each molecule has the same probability of being in the left half of the box as in the right half. The arrangement in

(a) corresponds to configuration III in **Table 20-1**

(b) corresponds to configuration IV.

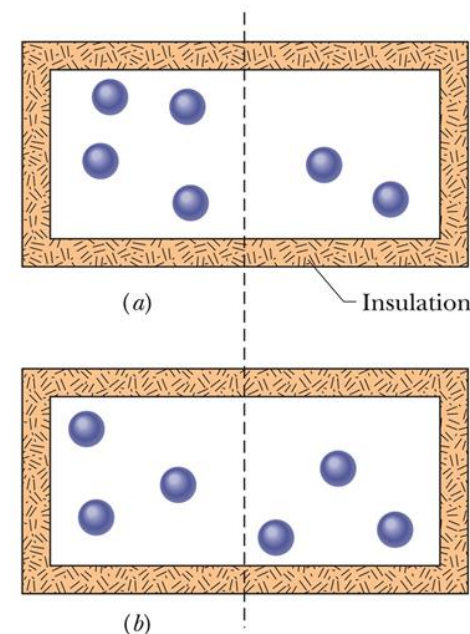


Table 20-1 Six Molecules in a Box

Configur ation	Configurat ion	Configu ration	Multiplicity W (number of microstates)	Calculation of W (Eq. 20-20)	Entropy
Label	n_1	n_2			
I	6	0	1	$\frac{6!}{(6!0!)} = 1$	0
II	5	1	6	$\frac{6!}{(5!1!)} = 6$	2.47
III	4	2	15	$\frac{6!}{(4!2!)} = 15$	3.74

Configur ation	Configurat ion	Configu ration	Multiplicity W (number of microstates)	Calculation of W (Equation 20-20)	Entropy 10^{-23} J/K (Equation 20 - 21)
Label	n_1	n_2			
IV	3	3	20	$\frac{6!}{(3!3!)} = 20$	4.13
V	2	4	15	$\frac{6!}{(2!4!)} = 15$	3.74
VI	1	5	6	$\frac{6!}{(1!5!)} = 6$	2.47
VII	0	6	1	$\frac{6!}{(0!6!)} = 1$	0

Total = 64

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Probability and Entropy

The multiplicity W of a configuration of a system and the entropy S of the system in that configuration are related by **Boltzmann's entropy equation**:

$$S = k \ln W \quad (\text{Boltzmann's entropy equation}).$$

Here $k = 1.38 \times 10^{-23} \text{ J/K}$ is the Boltzmann constant.

When N is very large (the usual case), we can approximate $\ln N!$ with **Stirling's approximation**:

$$\ln N! \approx N (\ln N) - N \quad (\text{Stirling's approximation}).$$

Order to Disorder

Entropy is a measure of the disorder of a system. This gives us yet another statement of the second law:

Natural processes tend to move toward a state of greater disorder.

Example: If you put milk and sugar in your coffee and stir it, you wind up with coffee that is uniformly milky and sweet. No amount of stirring will get the milk and sugar to come back out of solution.

Unavailability of Energy; Heat Death

Another consequence of the second law:

In any natural process, some energy becomes unavailable to do useful work.

If we look at the universe as a whole, it seems inevitable that, as more and more energy is converted to unavailable forms, the ability to do work anywhere will gradually vanish. This is called the heat death of the universe.

Thermodynamic Temperature; Third Law of Thermodynamics

Since the ratio of heats exchanged between the hot and cold reservoirs in a Carnot engine is equal to the ratio of temperatures, we can define a temperature scale using the triple point of water:

$$T = (273.16\text{K})(Q/Q_{\text{tp}}).$$

Here, Q and Q_{tp} are the heats exchanged by a Carnot engine with reservoirs at temperatures T and T_{tp} .

Also, since the maximum efficiency of a heat engine is

$$\varepsilon_C = 1 - \frac{T_L}{T_H}$$

there is no way to achieve a temperature of absolute zero. This is the third law of thermodynamics:

It is not possible to reach absolute zero in any finite number of processes.

Over 90% of the energy used in the U.S. is generated using heat engines to drive turbines and generators—even nuclear power plants use the energy generated from fission heat water for a steam engine. The thermal output Q_L of all these heat engines contributes to warming of the atmosphere and water. This is an inevitable consequence of the second law of thermodynamics.



Learning Outcomes

- Explain the concept of entropy and calculate its value.
- Analyze how a typical heat engine works.
- Analyze how a typical refrigerator works.
- Explain the 2nd (statistic) and 3rd laws of thermal dynamics.

$$\Delta S = S_f - S_i = \int_i^f \frac{dQ}{T} \quad S = k \ln W \quad W = \frac{N!}{n_1! n_2!} \quad \Delta S \geq 0$$

$$\varepsilon = \frac{\text{energy we get}}{\text{energy we pay for}} = \frac{|W|}{|Q_H|} \quad \varepsilon_C = 1 - \frac{|Q_L|}{|Q_H|} = 1 - \frac{T_L}{T_H}$$

$$K = \frac{\text{what we want}}{\text{what we pay for}} = \frac{|Q_L|}{|W|} \quad K_C = \frac{|Q_L|}{|Q_H| - |Q_L|} = \frac{T_L}{T_H - T_L}$$

CHAPTER REVIEW AND EXAMPLES, DEMOS

<https://openstax.org/books/university-physics-volume-2/pages/4-summary>

A **reversible process** is a process in which the system and environment can be restored to exactly the same initial states that they were in before the process occurred, if we go backward along the path of the process. The necessary condition for a reversible process is therefore the **quasi-static** requirement. Note that it is quite easy to restore a system to its original state; the hard part is to have its environment restored to its original state at the same time.

at any given moment of the process, the system most likely is not at equilibrium or in a well-defined state. This phenomenon is called **irreversibility**.

The effectiveness or **coefficient of performance** K_R of a refrigerator is measured by the heat removed from the cold reservoir divided by the work done by the working substance cycle by cycle:

$$K_R = \frac{Q_c}{W} = \frac{Q_c}{Q_h - Q_c}. \quad (4.3)$$

Note that we have used the condition of energy conservation, $W = Q_h - Q_c$, in the final step of this expression.

The effectiveness or coefficient of performance K_P of a heat pump is measured by the heat dumped to the hot reservoir divided by the work done to the engine on the working substance cycle by cycle:

$$K_P = \frac{Q_h}{W} = \frac{Q_h}{Q_h - Q_c}. \quad (4.4)$$

Example 4.3

A Carnot Heat Pump

Imagine a Carnot heat pump operates between an outside temperature of $0\text{ }^{\circ}\text{C}$ and an inside temperature of $20.0\text{ }^{\circ}\text{C}$. What is the work needed if the heat delivered to the inside of the house is 30.0 kJ ?

Example 4.6

Stirling Engine

The steps of a reversible Stirling engine are as follows. For this problem, we will use 0.0010 mol of a monatomic gas that starts at a temperature of $133\text{ }^{\circ}\text{C}$ and a volume of 0.10 m^3 , which will be called point A. Then it goes through the following steps:

1. Step *AB*: isothermal expansion at $133\text{ }^{\circ}\text{C}$ from 0.10 m^3 to 0.20 m^3
2. Step *BC*: isochoric cooling to $33\text{ }^{\circ}\text{C}$
3. Step *CD*: isothermal compression at $33\text{ }^{\circ}\text{C}$ from 0.20 m^3 to 0.10 m^3
4. Step *DA*: isochoric heating back to $133\text{ }^{\circ}\text{C}$ and 0.10 m^3

(a) Draw the pV diagram for the Stirling engine with proper labels.

(b) Fill in the following table.

Step	W (J)	Q (J)	ΔS (J/K)
Step <i>AB</i>			
Step <i>BC</i>			
Step <i>CD</i>			
Step <i>DA</i>			
Complete cycle			

(c) How does the efficiency of the Stirling engine compare to the Carnot engine working within the same two heat reservoirs?

You might suspect that the growth of different forms of life might be a net ordering process and therefore a violation of the second law. After all, a single cell gathers molecules and eventually becomes a highly structured organism, such as a human being. However, this ordering process is more than compensated for by the disordering of the rest of the universe. The net result is an increase in entropy and an increase in the disorder of the universe.

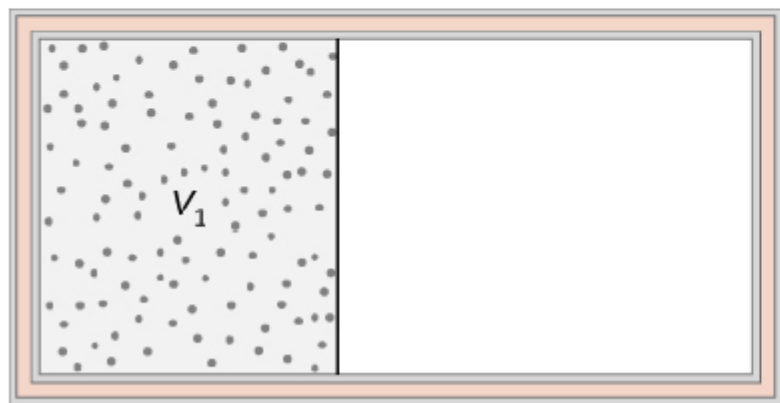
It turns out the absolute zero temperature is not reachable—at least, not though a finite number of cooling steps. This is a statement of the **third law of thermodynamics**, whose proof requires quantum mechanics that we do not present here.

(next example) The adiabatic free expansion of an ideal gas is an irreversible process. There is no change in the internal energy (and hence temperature) of the gas in such an expansion because no work or heat transfer has happened. Thus, a convenient reversible path connecting the same two equilibrium states is a slow, isothermal expansion from V_1 to V_2 . In this process, the gas could be expanding against a piston while in thermal contact with a heat reservoir, as in step 1 of the Carnot cycle.

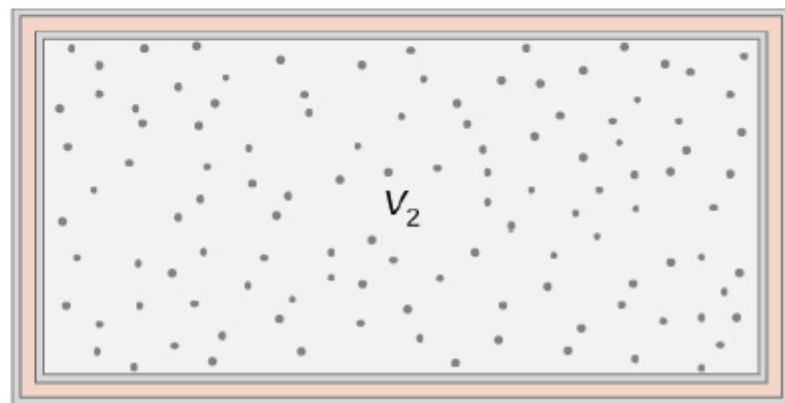
Example 4.8

Entropy Change of an Ideal Gas in Free Expansion

An ideal gas occupies a partitioned volume V_1 inside a box whose walls are thermally insulating, as shown in **Figure 4.18(a)**. When the partition is removed, the gas expands and fills the entire volume V_2 of the box, as shown in part (b). What is the entropy change of the universe (the system plus its environment)?



(a)



(b)

Figure 4.18 The adiabatic free expansion of an ideal gas from volume V_1 to volume V_2 .

Questions

